FPGA On-Board Demo and Test Flow

1. Purpose

This document explains how to run the FPGA on-board demo for the **RISC-V RV32 CPU with a Memory-Mapped CNN Coprocessor** final project.

The FPGA demo is controlled by three user inputs:

- **tc[3:0]:** selects which testcase to run or display.
- **mode:** selects the display mode.
- **st:** starts the selected testcase when mode = 0.

The result is shown on the four active seven-segment digits.

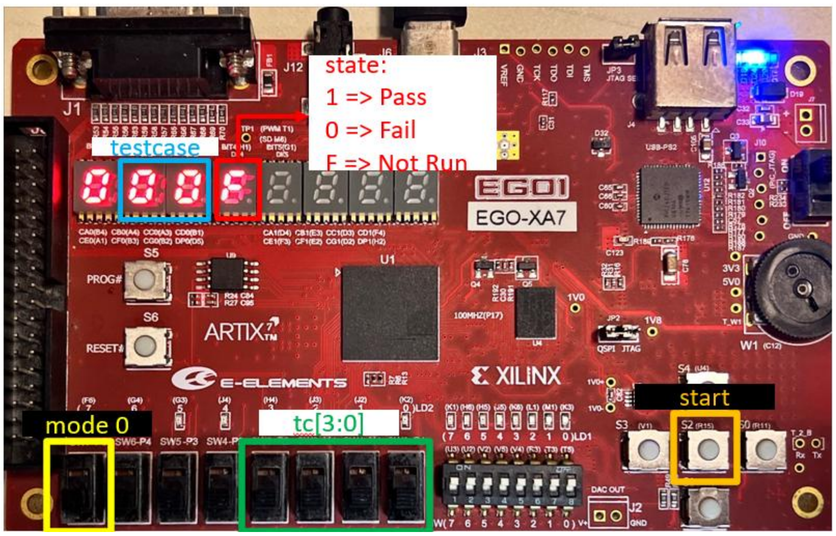

Fig1. Overall FPGA pins(mode = 0)

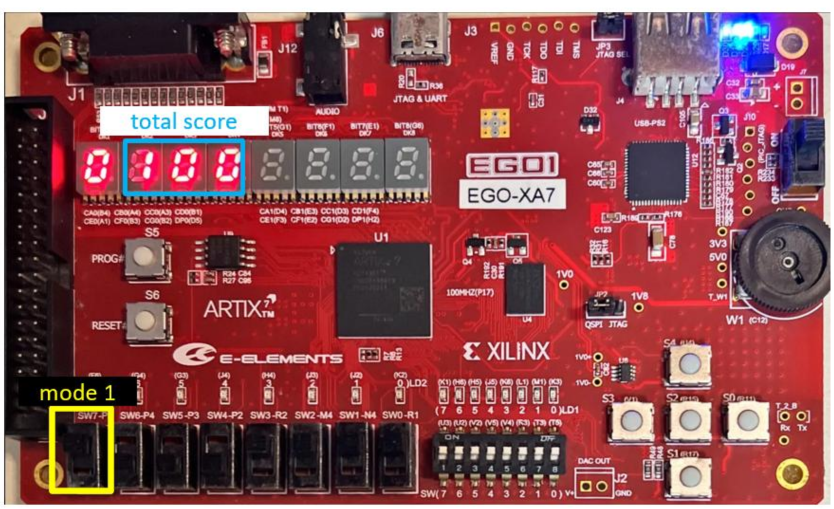

Fig2. Overall FPGA pins(mode = 1)

2. Testcase Selection

The input tc[3:0] selects the testcase. The mapping is shown below.

tc value	Test target	Description
0~9	CNN testcase 0~9	Feature map size 10~32
10	CPU testcase 0	addi / sw correctness
11	CPU testcase 1	lw correctness
12	CPU testcase 2	add / sub correctness
13	CPU testcase 3	beq correctness
14	CPU testcase 4	blt correctness
15	Run all / summary	Runs all testcases and displays total passed count

3. Display Modes

The mode input controls what the seven-segment display means.

mode	Function	How to interpret the display
0	Run testcase mode / testcase result mode	Use tc to select a testcase. Press st to start testing. The display shows the selected testcase number and its pass/fail status.
1	Score display mode	The display shows the current score scaled to 100 points.

4. Meaning of the Seven-Segment Display

4.1 Individual Testcase Display (mode = 0, tc = 0 to 14)

When mode = 0 and tc is between 0 and 14, the display format is:

TTTS

where:

- **TTT** is the selected testcase number in decimal.
- **S** is the testcase status.

The status digit has the following meaning:

Status digit	Meaning
F	The selected testcase **has not been executed yet**.
1	The selected testcase has **passed**.
0	The selected testcase has **failed**.

For example:

000F

means testcase 0 has not been executed yet.

0001

means testcase 0 has passed.

4.2 Run-All Summary Display (mode = 0, tc = 15)

When **tc = 15**, the test circuit runs all testcases. After the run-all process finishes, the display shows the total number of passed testcases.

For example:

0015

means all 15 test items passed.

Before pressing st, the summary display may show:

0000

which means no run-all result has been generated yet.

4.3 Score Display (mode = 1)

When mode = 1, the display shows the current functionality score scaled to 100 points.

The score is computed from the passed test items:

- **10 CNN testcases:** each passed CNN testcase contributes 5 points.
- **5 CPU subtests:** each passed CPU subtest contributes 10 points.

Therefore, the maximum score is:

10 CNN tests × 5 points + 5 CPU tests × 10 points = 100 points

5. Basic Demo Flow

Follow the steps below to run an individual testcase on the FPGA board.

Step 1: Reset the board

Press the reset button to initialize the system before starting the demo.

Step 2: Select run mode

Set mode = 0

This enables testcase execution and testcase result display.

Step 3: Select a testcase

Set tc[3:0] to the testcase that you want to run.

For example, to run CNN testcase 0:

tc = 0

Before pressing the start button, the display should show that this testcase has not been executed yet.

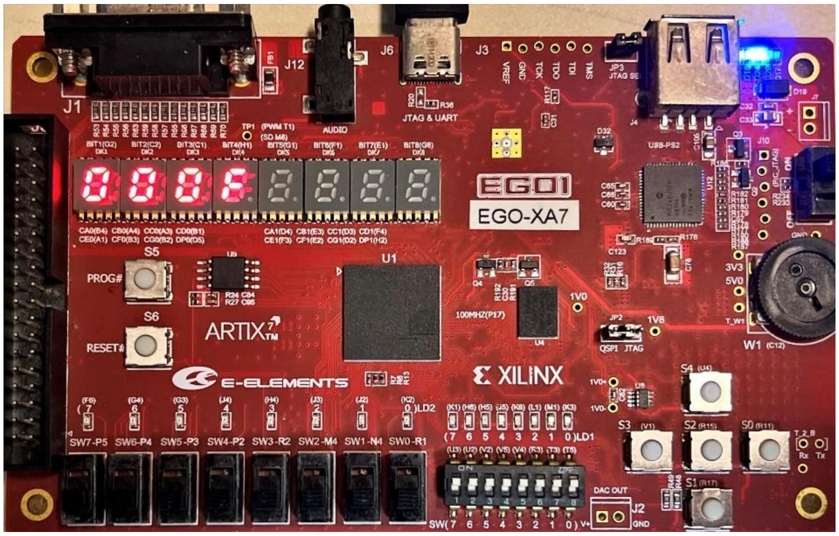

Fig3. Testcase 0 before pressing start: display shows 000F

In this example, the display shows 000F, which means testcase 0 is selected and its result is not available yet.

Step 4: Press the start button

Press st once to start the selected testcase.

Step 5: Check the result

After the selected testcase finishes, keep mode = 0 and check the status digit.

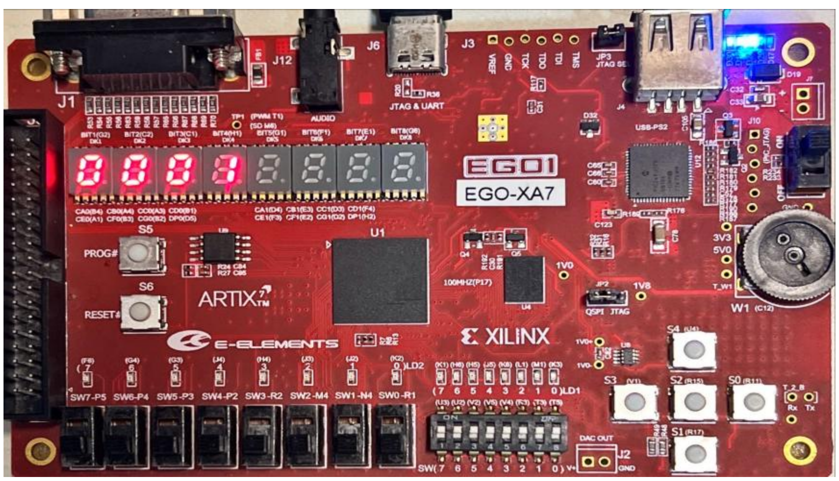

Fig4. Testcase 0 after pressing start: display shows 0001

In this example, the display shows 0001, which means testcase 0 has passed.

If the testcase fails, the last digit will show 0 instead of 1.

6. Viewing the Score of the Current Test Results

After running one or more testcases, set:

mode = 1

The display will show the current accumulated score.

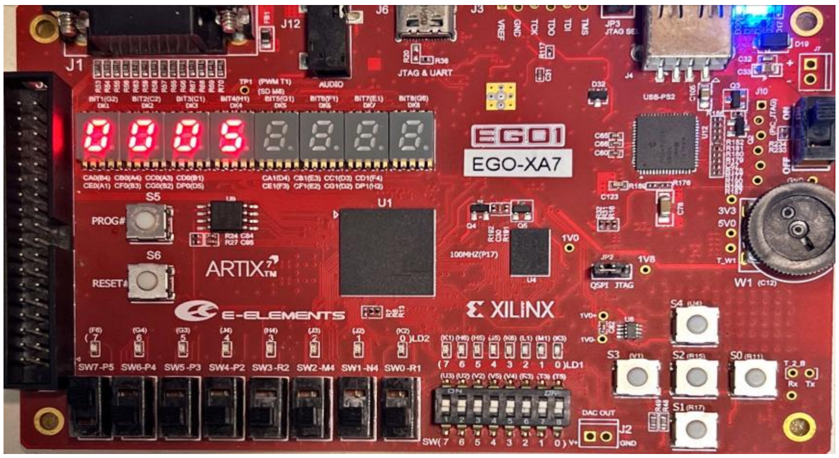

Fig5. Mode 1 after testcase 0: display shows the current score

In this example, only testcase 0 has passed, so the score display shows the score contributed by the currently passed tests.

The score display can also be used after running all testcases.

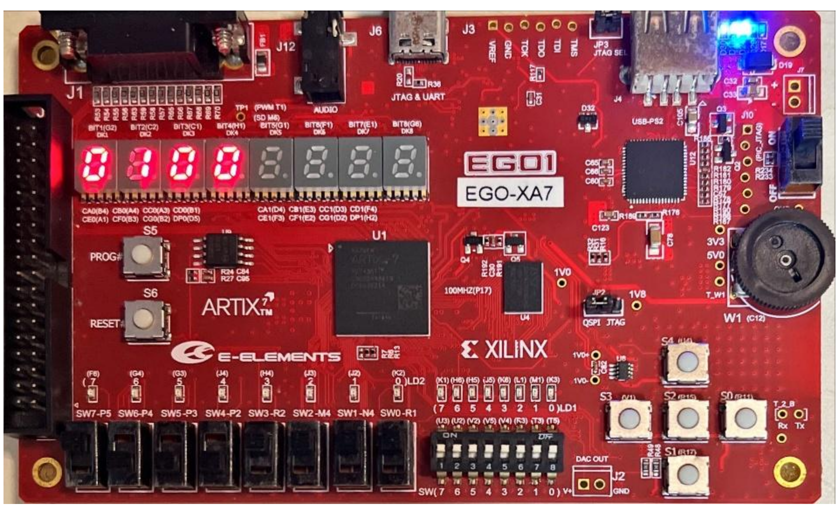

Fig6. Mode 1 score display after running all testcases

In this example, the display shows 0100, which means the current functionality score is 100 out of 100.

7. Running All Testcases

To run the complete on-board test, set:

mode = 0
tc = 15

Before pressing st, the display may show all zeros because no complete run-all result has been generated yet.

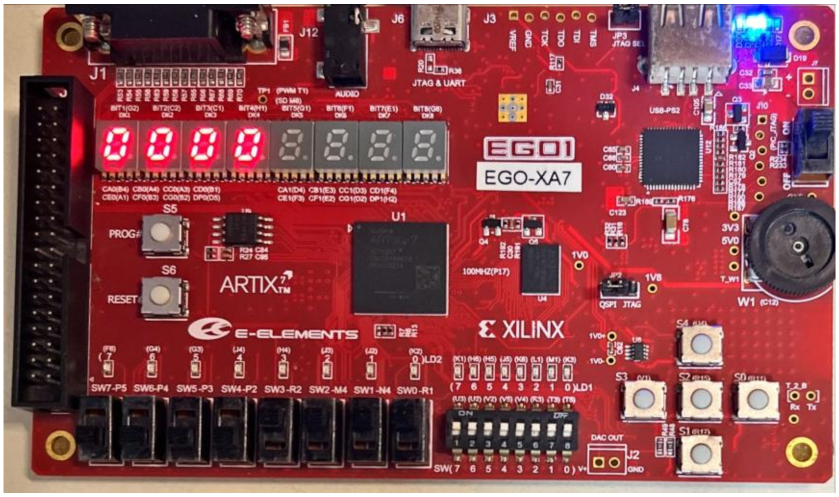

Fig7. tc = 15 before pressing start: display shows 0000

Then press st once.

The test circuit will run all 10 CNN testcases and all 5 CPU subtests. After the run-all process finishes, the display shows the total number of passed tests.

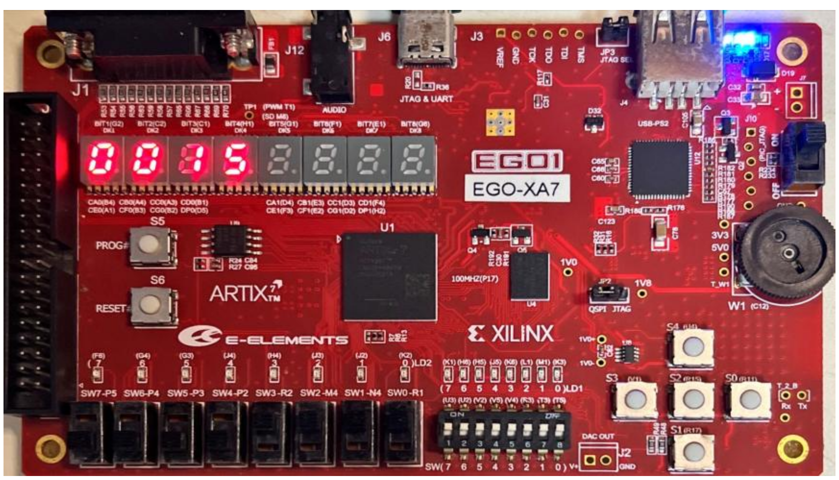

Fig8. tc = 15 after pressing start: display shows 0015

In this example, the display shows 0015, meaning that all 15 test items passed.

After this step, switching to mode = 1 should display the final score. If all test items passed, the score should be 0100.

8. Recommended Student Demo Procedure

During the demo, students are recommended to follow this order:

1. Reset the FPGA board.
2. Set mode = 0.
3. Set tc = 15.
4. Confirm the display is initially 0000 or an inactive summary value.
5. Press st once.
6. Wait until the run-all test completes.
7. Check whether the display shows 0015.
8. Switch to mode = 1.
9. Check whether the score display shows 0100.
10. If the total result is not full score, switch back to mode = 0 and test individual tc values from 0 to 14 to identify which testcase failed.

9. Debugging Individual Testcases

If the run-all result is not 0015, use the following method to locate the failed testcase.

1. Set mode = 0.
2. Select one testcase using tc.
3. Press st.
4. Check the last digit of the display.

Use the following interpretation:

Display pattern	Meaning
0001	CNN testcase 0 passed.
0000	CNN testcase 0 failed.
0101	CNN testcase 10 passed.
0100	CNN testcase 10 failed.
0141	CPU subtest 14 passed.
0140	CPU subtest 14 failed.

For CPU-related debugging, use the following subtest mapping:

tc	CPU function being checked
10	addi / sw
11	lw
12	add / sub
13	beq
14	blt

For CNN-related debugging, testcases 0 to 9 correspond to different feature map sizes. If only larger testcases fail, the problem may be related to memory addressing, output feature map size calculation, or loop boundary handling.